Crowdsensed Drive Test

Group 6

Introduction

- This project focuses on developing a Crowdsensed Drive Test Application.
- The system utilizes mobile crowdsensing to collect network performance data directly from user devices.
- Aim: To create high-resolution predictive maps of mobile network coverage.

Focus

- Identify network coverage needs.
- Understand user connectivity challenges.
- Define technical system requirements.

Key Stakeholders

Primary Stakeholders

Mobile Users:

- Main data providers.
- Need reliable and consistent signal coverage.

Network Operators (MNOs):

- Utilize data to justify infrastructure upgrades.
- Track real-world network performance.

Secondary Stakeholders

Project Developers:

- Build and maintain the sensing platform.

Telecommunications Regulatory Boards:

- Monitor service quality standards.

Infrastructure Providers:

- Affect signal quality through physical tower placement.

Requirement gathering techniques

Techniques:

- **Surveys**
- **Interview**
- **Brainstorming**

Surveys : forms distributed for data collection

Interview: on-sight interview with people to get their response

Brainstorming: analyzing raw data and responses from users to come out with the requirements for the system

System Requirements

Functional Requirements:
- Real-time data collection
- Coverage prediction
- Live signal visualization
- User notifications

Non-Functional Requirements:
- Privacy protection
- Low battery usage
- Scalability
- Fast performance

Data Collection

- **Surveys:** 65 responses
- **Interview :** 10 responses
- **Questions :** 20 questions
- **Format :** .xlsx , .pdf , .cvs

Data analysis

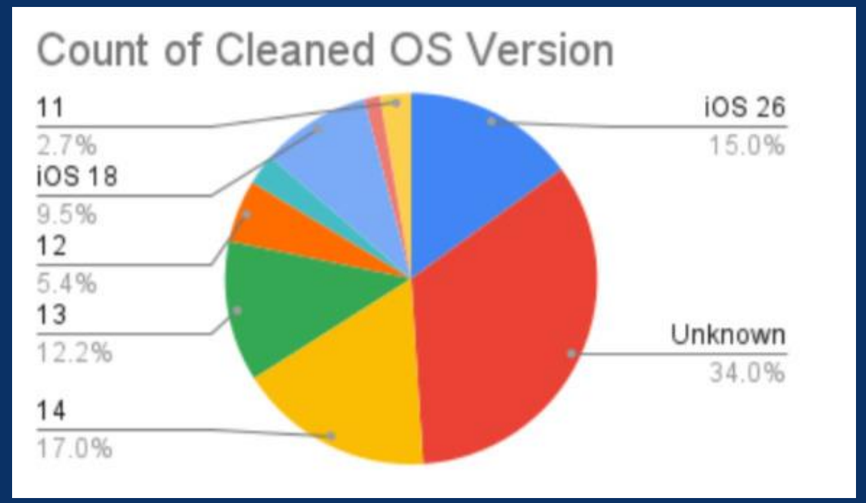
- **Standardization of OS Versions:** Many responses contained parenthetical

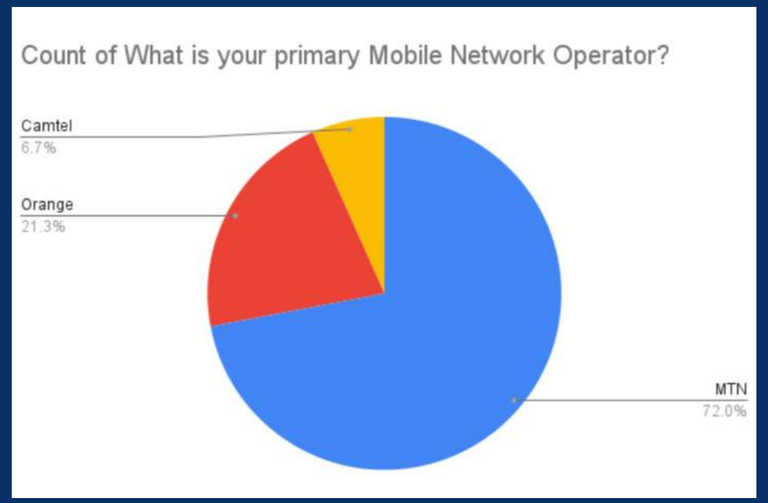
- **Normalization of Network Operators:** User entries for ISPs were inconsistent

- **Handling "Unknown" Responses:** A significant number of participants

- **Validation:** All 75 rows were reviewed for duplicate or "test" entries. The final dataset consists of 75 unique, valid data points.

Chart response





Data Cleaning

Our data cleaning process involved three key steps:

• **Standardization:** Unified formats for device names, OS versions, and network providers across all 75 survey responses

• **Normalization**: Converted all data to consistent scales and categories for accurate comparison

• **Validation:** Cross-checked responses to remove duplicates and fix errors

Tools Used: Google Forms (collection), Google Sheets (cleaning), Excel (analysis)

Result: Clean, accurate dataset ready for insights

User Reluctance

Platform Insights: Android dominates with 61% of users. This means our app must be Android-first. Most devices run recent OS versions, making modern features accessible.

Network & Performance: MTN is the leading provider in Buea. Users prioritize battery life and app performance. The app must be lightweight and power-efficient to succeed.

User Concerns

Privacy: Users worry about data collection and tracking. Many fear their location and usage patterns could be misused or shared without consent.
Complexity: Apps with too many features or confusing interfaces discourage adoption. Users prefer simple, intuitive designs.

Performance: Heavy apps drain battery and slow devices. Users avoid apps that consume too many resources.
Trust: New apps face skepticism. Users need proof that the app is safe, reliable, and beneficial before downloading.

Thank You !